DELHI PUBLIC SCHOOL NEWTOWN
SESSION 2025-26
HALF YEARLY EXAMINATION

CLASS: X | **FULL MARKS: 100**
SUBJECT: COMPUTER APPLICATIONS [SET A] | **TIME: 2 HOURS**

Answers to this Paper must be written on the paper provided separately.
You will not be allowed to write during the first 15 minutes.
This time is to be spent in reading the question paper.
The time given at the head of this Paper is the time allowed for writing the answers.

This paper consists of six printed pages. This Paper is divided into two Sections.
Attempt all questions from Section A and any four questions from Section B.
The intended marks for questions or parts of questions are given in brackets [].

SECTION A(40 Marks)

(Attempt all questions from this Section)

Question 1 **[20x1=20]**

Choose the correct answers to the questions from the given options

(i) 

Identify the feature of Java depicted by the above picture:
(a) Encapsulation (b) Data Abstraction (c) Inheritance (d) Polymorphism

(ii) **__ str="6734.3421".indexOf('5');**
Select the data type and the value stored in str:
(a) String and 0 (b) int and 0 (c) String and -1 (d) int and -1

(iii) Choose the correct output:
System.out.println(**Math.pow(Math.min(Math.sqrt(16),Math.cbrt(9))),3**));
(a) 9.0 (b) 9 (c) 27.0 (d) 49

(iv) A char primitive type consumes ________ of memory.
(a) 2bits (b)4bits (c) 16bits (d) 8bits

(v) Choose the correct output:
System.out.println(**String.valueOf(2589).charAt(2));**
(a) Error (b) Null (c) 5 (d) 8

(vi) Method prototype that accepts two integer and returns a character:
(a) int fun(int a, int b) (c) char fun(int a,b)
(b) char fun(int a, int b) (d) String fun(int a, int b)

(vii) Choose the correct output:
System.out.println(**Double.parseDouble("134")+'a');**
(a) 231.0 (b)Error (c)199 (d) “134a”

(viii) Choose the correct output if **n=3**:

```
static void fun1(int n) {
int i, f;
for(i=1,f=1;i<=n;i++,f*=i){ }
System.out.print(f); }
```

(a) 120 (b) Error (c)1 (d) 24

(ix) Choose the correct output:

```
String a1="Joyful", b1="Joyfully";
System.out.println(a1.substring(4, a1.length()).equals(b1.substring(6)));
```

(a) ful (b) Joy (c)true (d)false

(x) Assertion(A): An infinite loop is an instruction sequence that loops endlessly.
Reason(R): In the infinite loop the test condition is always true.
(a) Both Assertion (A) and Reason (R) are true and (R) is a correct explanation of (A)
(b) Both Assertion (A) and Reason (R) are true and (R) is not a correct explanation of (A)
(c) Assertion (A) is true and Reason (R) is false
(d) Assertion (A) is false and Reason(R) is true

(xi) Which of the following is not a Class in java library:
(a) Integer (b) Math (c) String (d) boolean

(xii) What will be the result of "Java".compareTo("Java")?
(a) 1 (b) 0 (c) -1 (d) True

(xiii) Assertion(A): String is Immutable.
Reason(R): Any operations that appears to modify a String actually creates a new String object.
(a) Both Assertion (A) and Reason (R) are true and (R) is a correct explanation of (A)
(b) Both Assertion (A) and Reason (R) are true and (R) is not a correct explanation of (A)
(c) Assertion (A) is true and Reason (R) is false
(d) Assertion (A) is false and Reason(R) is true

(xiv) Technique using which a class acquire the properties (fields) and behaviors (methods) of another class.
(a) Encapsulation (b) Data Abstraction (c) Inheritance (d) Polymorphism

(xv) Method Overloading is form of ____________________
(a) Encapsulation (b) Package (c) Inheritance (d) Polymorphism

(xvi) A method that uses static variables:
(a) void (b) return (c) non-static (d) static

(xvii) Calling a method should be done with the help of
(a) constructor (b) method prototype (c) method signature (d) loop

(*xviii*) Identify the class variable
(a) final int a=5 ; (b) public int a=5 ; (c) static int a=5 ; (d) int a=5;

(xix) The value returned by **Math.abs(-321.453));**
(a) 321.0 (b) -321.0 (c) 321.453 (d) “321”

(xx) A method that is same as class name and initializes by a value that is passed as a method arguement:
(a) Default Constructor (b) Copy Constructor
(c) Parameterized Constructor (d) Constructor Overloading

Question 2

i. Write the Java expression for **(–b+√(b^2-4ac))/2a**. [2]

ii. Evaluate the expression: **p*=p-- + ++p/p + p--; where p=5;** [2]

iii. Give the output of the following code-snippet: [2]

```
String a="42431";
System.out.println(Math.pow(a.length(),a.indexOf('2')));
```

iv. Give the output of the following code-snippet: [4]

```
String x1 = "  Trim this string!  ";
System.out.println(">" + x1.trim() + "<");
System.out.println(x1.replace(' ', '_'));
System.out.println(x1.indexOf('t',7 ));
System.out.println(x1.substring(14,x1.length()-3));
```

v. Write the return type of the following library functions: [2]
  a) isLetterOrDigit(char)
  b) replace(char, char)

vi. Rewrite the following code using the while loop: [2]

```
int s,a;
for(s=1,a=12;a>=3;a - =2)
s*=a;
System.out.println(“Result=”+s);
```

vii. Write the output of the following program code: [2]

```
char ch ;    int x=97;
do{    ch=(char)x;
System.out.print(ch + " " );
if(x%10 == 0)    break;
++x;   } while(x<=100);
```

viii. Write a statement each to perform the following tasks on a string: [2]
a) Find and display the position of the last space in a string s.
b) Convert a number stored in a string variable x to integer data type.

ix. Give differences between break and continue with the help of an example. [2]

SECTION B (60 Marks)

(Answer any four questions from this Section)

The answers in this section should consist of the programs in either BlueJ environment or any program environment with java as the base. Each program should be written using variable description/mnemonic codes so that the logic of the program is clearly depicted. Flowcharts and algorithms are not required.

Question 3 [15]

Define a class named BookFair with the following description:

Class Name : BookFair

Member variables :

String Bname : to store the name of the book

double price : to store the price of the book

double bill : to store the bill amount

Member functions

BookFair() : Default constructor to initialize data members

void accept( ) : to accept the name and price of the book

void calculate() :

Price	Discount
Less than or equal to ₹1000	2% of price
More than ₹1000 and less than or equal to ₹3000	10% of price
More than ₹3000	15% of price

void print( ) : To display the name and price of the book after discount in the given format:

Book Name	Price	Bill Amount
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Define a class with the above mentioned specifications, create the main method, create an object and invoke the member methods.

Question 4 [15]

Write a program to **accept two numbers** from the user and check whether they are **co-prime or not.**

Two integers a and b are said to be relatively prime, mutually prime, or co-prime if the both the entered numbers are prime or the only positive integer that divides both of them is 1.

Example: 13 and 15 are co-prime.

Question 5 [15]

Design a class to **overload a function show( )** as follows:

void show(String s): to accept a sentence and display the total number of words in the given sentence.

Example :

Input:

s= “Have a nice day”

Output:

Total number of words=4

void show( ): dipslay the following pattern.

225	196	169	144	121
100	81	64	49	
36	25	16		
9	4			
1				

Question 6 [15]

Write a program to accept a string. Convert the string into upper case letters. Count the number of **double letter sequences** that exist in the string and display the resultant string after **replacing the double letters with single letter**.

Sample Input: "SHE WAS FEEDING THE LITTLE RABBIT WITH AN APPLE"

Sample Output:

Total number of double letters word=4

Resultant String= SHE WAS FEDING THE LITLE RABIT WITH AN APLE"

Question 7 [15]

Design a class to **overload a function count( )** as follows:

int count(int a, int b): to accept a range(a and b, where a<b and a, b are exclusive) from the user and count and return the total number of DivNine numbers present in the given range.
A DivNine number is either divisible by 9 or startswith 9. Eg: 81 and 91.

int count (String s, char c): to accept a word and a character and count the frequency of the character in the given string.
Example :
Input:
s= “Hello”;
c= ‘l’;
Output:
Frequency =2
[Do not write the main method.]

Question 8 [15]

Write a program to accept a sentence from the user. Display all the **palindromes** and the **words starting and ending with vowels** present in the given sentence.

Example:

Input: “Eve admired an aura, radar echo, civic rotor alive under an aurora.”

Output: Palindromes: Eve, radar, civic, rotor

Words starting and ending with vowels: admired, aura ,echo, alive, aurora